

Analog control systems

- Mathematical modeling
- Geometric root locus analysis
- Frequency response analysis
- P, PI and PID compensators

Microcontrolled systems

- Analog-to-digital converter
- Pulse-width modulation (PWM)
- Interrupt routines
- P, PI and PID compensators

Digital control systems

- Discrete mathematical modeling
- Sampling theorem
- Emulation of discrete compensator
- Discretization techniques

Digital signal processing

- Signal processing
- Digital filters (Moving average, FIR, IIR)

Physics

- Voltage, current and electrical power
- Ohm and Kirchoff's law

Electronics and instrumentation

- Rectifiers circuits
- Signal conditioning circuit
- Zero-crossing detection circuit
- Drive circuit for alternating current loads
- Analog filters

Introduction to engineering

- Understanding of applied projects
- Practical application of voltage conversion systems